Football Tactical Analysis System with YOLO and LLM

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***Abstract*—Analyzing football tactics from video traditionally requires labor-intensive manual review by coaches and analysts. To address this, a novel real-time system is proposed that combines computer vision and language modeling to automate tactical insights. The system utilizes a YOLOv8-based vision module to detect players and the ball, which are then tracked over time using DeepSORT. Image positions are projected to standard pitch coordinates via homography calibration to compute spatial metrics like heatmaps and defensive lines. Simultaneously, a football-specific Large Language Model (LLM) generates natural-language commentary to explain the observed tactics. Preliminary implementation demonstrates effective team classification using a color-based K-means classifier. This multimodal pipeline bridges the gap between raw video data and high-level tactical reasoning, providing both graphical and textual analyses for coaches and fans.**

***Index Terms*—YOLOv8, DeepSORT, Large Language Models (LLM), Tactical Overlays, Sports Analytics, Homography Projection**

I. Introduction

Football tactical analysis is a critical component for professional teams, yet manual video review is time-consuming. Recent advancements in Artificial Intelligence (AI) and Computer Vision (CV) have begun to reduce this workload. Modern object detectors can identify players, balls, and referees in broadcast footage in real-time. However, selecting appropriate algorithms to transform these detections into meaningful tactical insights remains a challenge.

While CV methods can extract positional data, they often lack the explanatory power required for deep tactical understanding. Recent benchmarks show that while LLMs handle basic sports facts, they struggle with complex scenario-based reasoning without domain-specific integration. This research introduces an end-to-end framework that unifies vision-driven analysis with LLM-based explanation. By integrating automated tactical overlays with natural-language commentary, the system provides a comprehensive diagnostic tool for match investigation.

II. MATERIALS AND METHODS

The proposed system architecture involves four successive procedures: (a) Detection and tracking of entities, (b) Projection of coordinates to pitch space, (c) Generation of tactical overlays, and (d) Automated commentary generation via LLM.

III. Process Architecture (End-to-End Workflow)

Fig. 2 illustrates the complete operational pipeline of the proposed system, starting from raw match footage and ending in multimodal tactical outputs. The input video stream is first decomposed into individual frames, which are processed by a real-time object detector to localize players, referees, and the ball. These detections are then passed to a multi-object tracking module, which assigns persistent identities and estimates short-term motion vectors.

Subsequently, higher-level tactical features such as team shape, pressing intensity, line compactness, and offside lines are computed in a normalized pitch coordinate system obtained via homography projection. These features are serialized into a structured tactical JSON representation, which serves as an intermediate, model-agnostic interface between the vision stack and the language module.

Finally, the structured representation is consumed by a football-specific Large Language Model (LLM), which performs contextual reasoning and generates both visual overlays and natural-language explanations. This modular design allows each stage of the pipeline to be independently improved or replaced without affecting the overall system.

A. Vision Module

The vision module processes each video frame to detect players and the ball using YOLOv8. To maintain consistent identities across frames, the DeepSORT tracker is employed. A specialized filter checks the lower region of detections for green field color to ensure entities are correctly situated on the pitch.

B. Team Classification and Localization

Team assignment is achieved using a lightweight K-means classifier on sampled jersey colors to separate players into Team A and Team B. Once identified, image coordinates are projected to a standard pitch coordinate system via homography calibration. This enables the computation of high-level features such as team heatmaps, defensive lines, and offside lines.

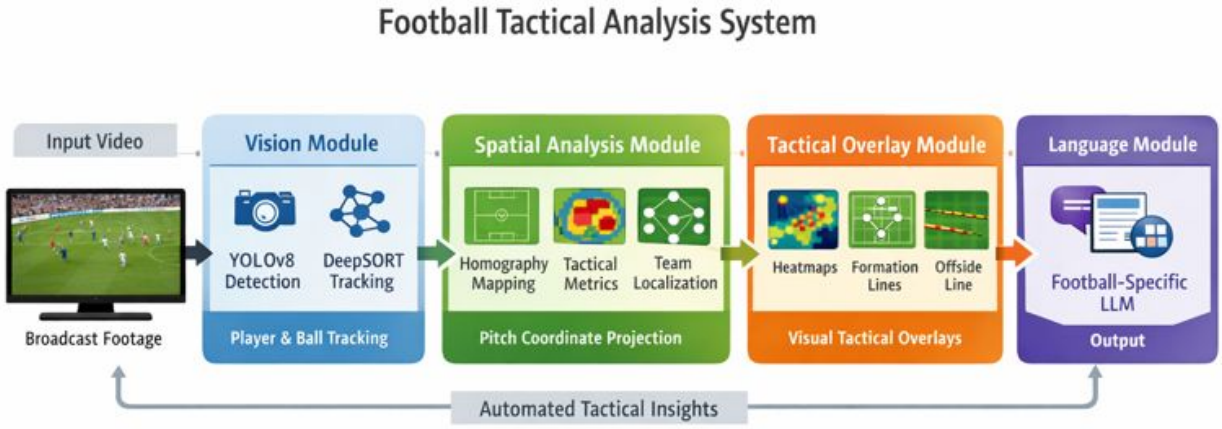


Fig. 1. Overall architecture of the proposed football tactical analysis system.

C. Language Module (LLM Integration)

1) Internal LLM Architecture: While the previous subsection describes the interface between the vision system and the language model, Fig. 6 details the internal structure of the LLM-based reasoning module. The input to this module is a structured tactical JSON containing normalized player positions, team-level metrics, and detected tactical patterns.

This representation is first processed by a prompt construction layer, which reformulates the raw numerical and categorical features into a coach-style contextual prompt. To improve domain consistency and reduce hallucinations, a retrieval-augmented generation (RAG) component is employed to fetch relevant tactical principles, such as pressing rules and shape constraints, from a curated knowledge base.

The resulting enriched prompt is then passed to a fine-tuned large language model (e.g., Mistral or LLaMA with LoRA adaptation), which has been adapted to the football analytics domain. The model outputs a structured tactical explanation and, when applicable, short-term tactical predictions. This design separates perception, knowledge retrieval, and reasoning, leading to more stable and interpretable textual analyses.

D. Language Module (LLM Integration)

E. Tactical Overlay Generator

Spatial data is rendered as visual overlays onto the video stream. These include:

Heatmaps: Highlighting high-activity zones.

Formation Lines: Inferring defensive and offensive structures.

Offside Indicators: Based on the position of the second-last defender.

F. Language Module (LLM Integration)

The system feeds a structured summary of the vision module's output—such as player positions and active tactical patterns—into a football-specific LLM. The LLM is prompted to generate insightful text, such as "Team Red is applying a high press," effectively linking visual cues to strategic reasoning.

IV. Model Selection and Justification

The choice of the object detection backbone is critical for real-time football analytics, where both accuracy and latency constraints must be satisfied. Traditional two-stage detectors such as Faster R-CNN provide strong accuracy but typically fail to meet real-time requirements, often operating below 20 FPS on standard GPU hardware. Transformer-based

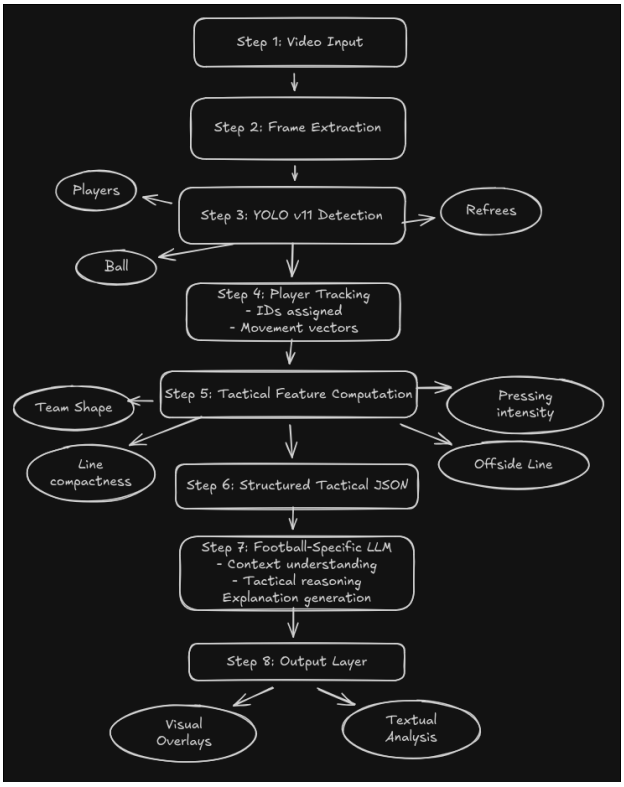


Fig. 2. End-to-end processing pipeline from raw match video to visual and textual tactical outputs.

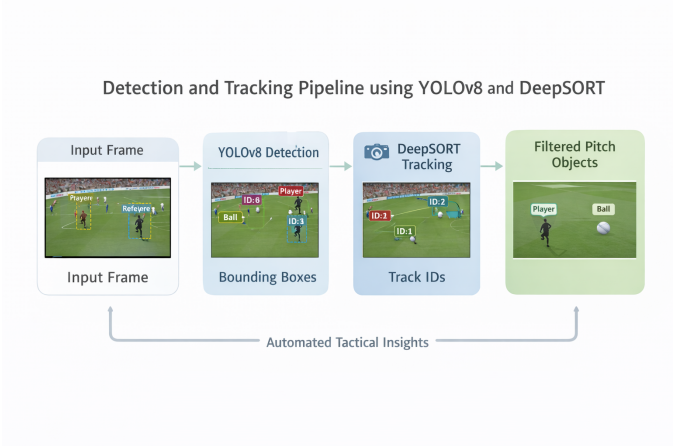


Fig. 3. Detection and tracking pipeline using YOLOv8 and DeepSORT.

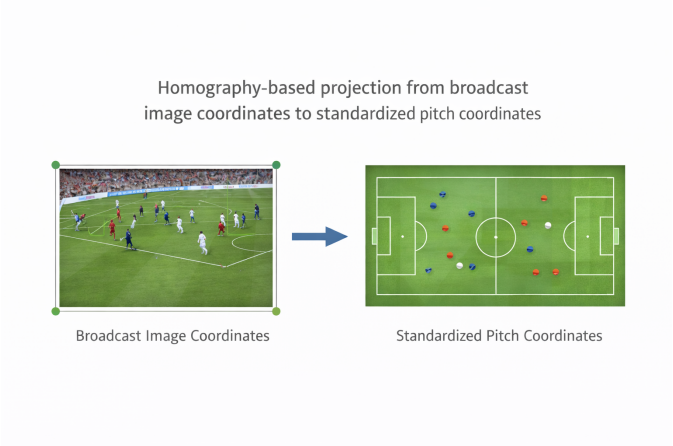


Fig. 4. Projection of image coordinates to standardized pitch coordinates using homography.

detectors such as RT-DETR improve global context modeling but introduce higher latency due to the quadratic complexity of attention mechanisms. In contrast, recent YOLO-family detectors offer a more favorable trade-off between speed and accuracy, making them well-suited for continuous video analysis.

Large-scale variants such as YOLOv8 and YOLO11 achieve state-of-the-art performance on the COCO Val2017 benchmark while maintaining real-time throughput on commodity GPUs, with YOLOv8-X reported at approximately 53.9% mAP and over 70 FPS on T4-class hardware. In this work, we employ a high-capacity YOLOv8/YOLO11-class model due to its stable deployment ecosystem and reproducible real-time performance. Although newer attention-centric architectures such as YOLOv12 further improve small-object recall, we prioritize robustness, tooling support, and ease of integration to ensure consistent latency and reliable operation in a production-oriented pipeline.

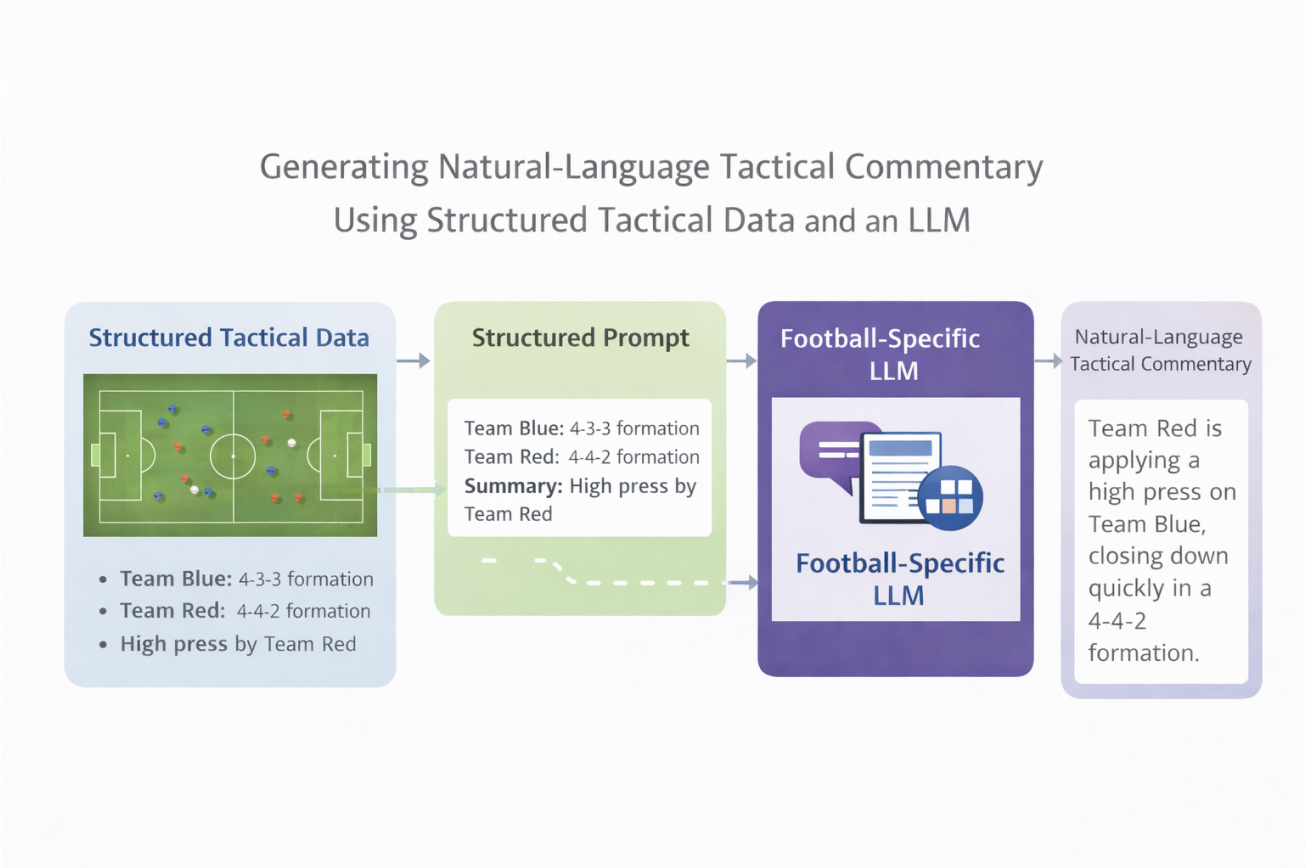


Fig. 5. Generated tactical overlays: (a) player heatmap, (b) defensive line estimation, (c) offside line detection.

V. EXPERIMENTS AND RESULTS

Preliminary implementation was performed using YOLOv8 for detection and a color-based K-means classifier for team assignment.

A. Detection and Tracking Performance

YOLO-style detectors were selected for their balance of speed and accuracy, essential for real-time analysis. DeepSORT allowed for efficient identity maintenance, which is critical for generating accurate player trajectories.

B. Team Assignment Accuracy

The K-means clustering approach on jersey-color histograms provided a robust baseline for separating competing teams. This automated classification is essential for calculating team-specific metrics like possession zones and average team positions.

C. Visual and Textual Output

The system successfully generated tactical graphics, such as semi-transparent heatmaps and defensive lines, overlaid on match footage. The integration of the LLM ensures that these visual patterns are not just seen but understood through synchronized textual commentary.

VI. CONCLUSION

This study provides an effective framework for automated football tactical analysis. By combining YOLO-based vision pipelines with explanatory LLMs, the system confirms that multimodal approaches can significantly enhance sports analytics. The duo approach of graphical overlays and natural-language insight reduces the manual burden on analysts while improving the depth of match interpretation. Future work will focus on refining formation inference and completing the full-scale fine-tuning of the LLM on expert coaching reports.

REFERENCES

[1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955.
[2] J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
[3] I. S. Jacobs and C. P. Bean, “Fine particles, thin films and exchange anisotropy,” in Magnetism, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.
[4] K. Elissa, “Title of paper if known,” unpublished.
[5] R. Nicole, “Title of paper with only first word capitalized,” J. Name Stand. Abbrev., in press.
[6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, “Electron spectroscopy studies on magneto-optical media and plastic substrate interface,” IEEE Transl. J. Magn. Japan, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetics Japan, p. 301, 1982].
[7] M. Young, The Technical Writer’s Handbook. Mill Valley, CA: University Science, 1989.

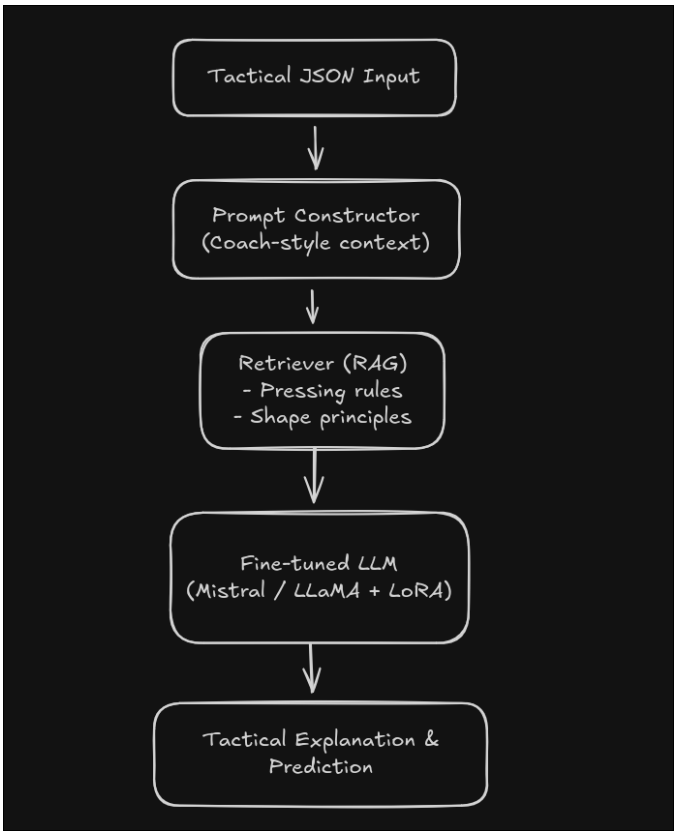


Fig. 6. Internal architecture of the LLM module showing prompt construction, retrieval-augmented generation, and fine-tuned model inference.

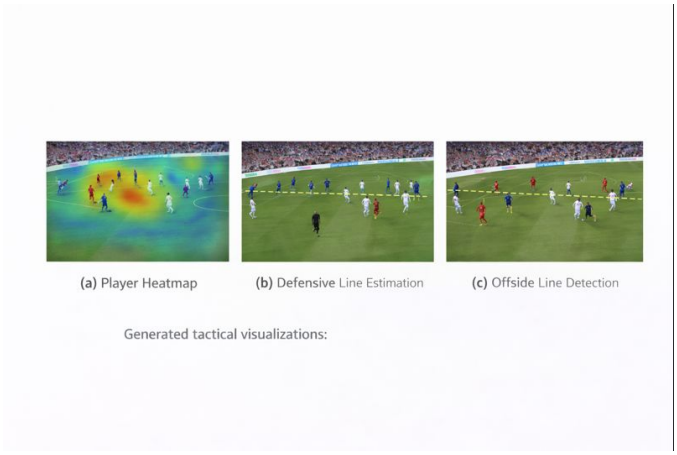


Fig. 7. Transformation of structured tactical data into natural-language explanations using an LLM.